Microeconomics

A project report
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Contents

0.1 Introduction

Mathematical principles unveil a world of intricacies within microeconomics which can be used to model markets. These relations shape the decisions and market dynamics of industries, firms, and consumers alike. At its core, the relationship between applied mathematics and microeconomics not only define, but also quantifies the fundamental forces governing market behaviors.

This report delves into many of the fundamental mathematical concepts which comprise microeconomic theory. From supply and demand to the calculations of revenue, cost, and profit, these concepts will define the basic theory behind microeconomics. Exploring the elasticity of demand becomes an integral part of this report; of which unravels the sensitivity of markets to fluctuations caused by various external factors. Furthermore, duopolistic competition will be explored with the purpose of offering insight into the decision-making process taken by market competitors, and how the consequences of the decisions effect market dynamics.

Lastly, the application of mathematical microeconomic models not only reveals the complexities within microeconomics itself, but also empowers businesses and consumers alike with tools to navigate the intricacies of market behavior. Through the lens of applied mathematics, this report offers a deeper understanding of the foundation upon which microeconomic theories stand.

0.2 Supply and Demand

Supply and demand are the forces that drive market economies. They represent the relationship between the availability of goods and services and the desire for them. When supply matches demand, markets reach equilibrium which provides stability to both the consumer and business.

The demand function itself is decreasing. We can first consider a demand function represented by

$$q = D(p) \tag{1}$$

where p is the price of a given item, and q is the quantity of units sold at p. Broadly speaking, higher values of p drive q to have a lower value, relative to the prices and quantities of a given market. This behaviour of *D(p)* will be further examined in this section.

In addition to the demand function, the supply function is an increasing function represented by

$$q = S(p) \tag{2}$$

where p is the price of a given item, and q is the quantity of units which are produced at price p. While *S(p)* is an increasing function, it should be noted that typically q is not positive until some threshold is reached.

The behaviour of these functions can highlight important details of a given unit subject to it's own supply and demand. For a given commodity, consider the following scenarios:

$$D(p) < S(p) \tag{3}$$

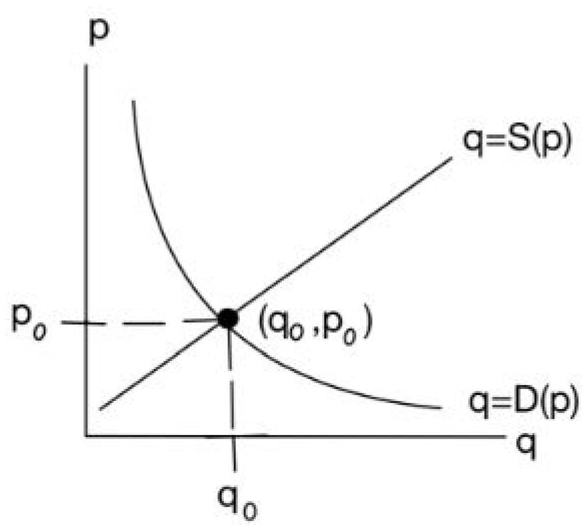
When the demand is lower than the supply, a price decrease occurs. This is defined as the *surplus* of a product. This highlights an abundance of goods available beyond what consumers seek or require.

Conversely, the opposite can occur:

$$D(p) > S(p) \tag{4}$$

When the demand is higher than the supply, a price increase occurs. This is defined as the *scarcity* of a product. In particular, this highlights an imbalance between consumer wants and the limited resources available to a producer.

When *S(p)* and *D(p)* are simultaneously graphed on the same x-axis, their intersection (q_0, p_0) yields an equilibrium price[1] . The graph below demonstrates this behaviour.



Supply: q=S(p)
Demand: q=D(p)
Equilibrium: (q,p)

Equilibrium price signifies the point where supply meets demand, establishing a stable market price. This is clearly shown at point (q_0, p_0) in the graph.

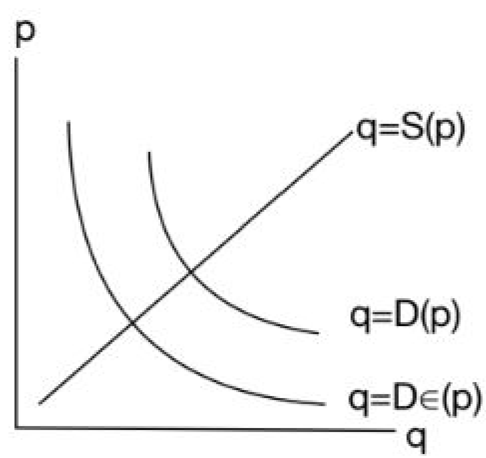
Thus far we have assumed the price (p) to be a price which was set by all producers. However, in reality there are many other factors that can effect price. To explore this concept, let's analyze the following question:

What is the effect of sales tax on price?

Recall that the demand function decreases as $p \to \infty$. Hence we can modify the demand function to account for the price increase modeled by:

$$q = D_\epsilon(p) = D(p + \epsilon p) \tag{5}$$

Due to p increasing by some factor of ϵ, where $\epsilon > 0$, this ultimately forces the demand function to lower its' value for q. Subsequently, the equilibrium price lowers. The graph below demonstrates how taxes and fees modify the equilibrium price as a whole.



While it is true that the public sector revenue, represented by $\epsilon \times p$, would be ever increasing as $p \to \infty$ and $\epsilon \to \infty$, there are further, practical implications to consider when increasing these values. If the public sector increases ϵ too high, the demand will fall, thus making the total revenue generated from $\epsilon \times p$ lower. Therefore, ϵ must slowly approach some *critical point* where the public sector can balance the revenue generated coupled with the subsequent change in demand.

0.3 Cost, Revenue, and Profit

In microeconomics, revenue signifies income from sales, while costs encompass production expenses. Profit emerges as the difference between revenue and costs, influencing firms' decisions on production, pricing, and resource management within competitive markets. A cost function is displayed below:

$$C = C(q) \tag{6}$$

Where q is the quantity produced from the demand, *D(p).* Traditionally cost tends to rise linearly [1]. However, as a company scales, they are hit with the inevitable issues which belong to producing mass amounts of goods. These problems include scheduling, labor shortages, supply shortages, commitments to clients, overtime, etc. As a result of these factors, $C(q)$ rises at an ever increasing rate; increasing faster as the amount of goods (q) increases.
Moving forward, profit is simply calculated by subtracting the revenue from the cost.

$$\pi(q) = R(q) - C(q) \tag{7}$$

While these formulas are the foundation of all models in microeconomics, businesses stand to derive more from these valuable metrics. Some of which are the marginal measurements. First, we can define *marginal cost* as the amount required to produce one more unit. We can approximate this by taking the difference between the cost of the current unit to produce, subtracted from the cost of the next unit to produce.

$$C(q+1) - C(q) \approx C'(q) \tag{8}$$

Marginal revenue is defined by the amount gained by selling one unit.

$$R'(q) \tag{9}$$

For business looking to maximize profit, they should determine when the marginal cost and marginal revenue equal zero.

$$R'(q) = C'(q) \tag{10}$$

To better illustrate these concepts, consider the following scenario: A business produces $5q$ units of revenue, while the cost is $q^2 + 4$. Given these measurements, how many units (in millions) should be produced to maximize profit?

$$R(q) = 5q, C(q) = q^2 + 4 \tag{11}$$

$$R'(q) = 5, \quad C'(q) = 2q \tag{12}$$

$$R'(q) = C'(q) \to 5 = 2q \to q = 2.5 \tag{13}$$

Thus, the business should produce 2.5 million units to maximize it's profit.

Generally speaking, maximizing revenue in microeconomics involves optimizing pricing strategies, production levels, and market positioning. These are all applications of linear programming. While this report does not focus on optimization and linear programming, having the foundation of microeconomics stem from linear functions provide desireable properties like simplicity, predictability, and interoperability. Balancing these elements enables firms to achieve optimal revenue generation, fostering competitiveness and sustainable growth within dynamic market landscapes.

0.4 Elasticity of Demand

Elasticity of demand, a pivotal concept in economics, quantifies the responsiveness of consumer demand to changes in price or other influencing factors [1]. Understanding demand elasticity provides vital insights into consumer behavior and market dynamics, guiding businesses and policymakers in pricing strategies, resource allocation, and forecasting outcomes within various economic landscapes.

It is desireable to quantify how a price increase from *(p)* to *(p + dp)* will affect demand $q = D(p)$. To understand this, it is necessary to understand the context of the price within a given market, coupled with how much the demand is. Consider the following statement:

"Raising our price by \$100 will decrease demand by 800 units."

Without the overall context of price and current demand, this statement is extremely ambiguous. How can someone determine the impact of this statement? The following are two scenarios which demonstrate the varying levels of impact this could produce. First, if the present demand is 900 units at a price of \$1000 per unit, the result is detrimental to the market and business. On the other hand, if the present demand is 32,000 units with a price of \$1000 per unit, the price increase would be beneficial to the market and business.

In applied mathematics, we can quantify this concept by introducing the *price elasticity of demand*. This is denoted by the letter e (not to be confused with Euler's number). It is defined by calculating the *percentage decrease in demand per percentage increase in price* [1].

$$e = -\frac{dq/q}{dp/p} = -\frac{dq}{dp} * \frac{p}{q} = -p * \frac{D'(p)}{D(p)} \tag{14}$$

If $e>1$, demand is considered to be elastic. This signifies that relatively small changes in price have a relatively large effect on the market and business. Alternatively, when $e <= 1$, demand is considered to be inelastic.

Generally speaking, this provides more flexibility to the business when making decisions and understand it's effect on the market.

The following demonstrates demand elasticity. Recall the scenario where a \$100 price increase with a present demand of 900 units, at a current price of \$1000 per unit. Thus e is calculated by:

$$e = \frac{dq/q}{dp/p} = \frac{(800/900)}{(100/1000)} = 8.9 > 1 \tag{15}$$

The following demonstrates demand inelasticity. Recall the scenario where a \$100 price increase with a present demand of 32,000 units, at a current price of \$1000 per unit; e is now calculated by:

$$e = \frac{dq/q}{dp/p} = \frac{(800/32,000)}{(100/1000)} = 0.25 < 1 \tag{16}$$

One of the key insights that the elasticity of demand brings to a business is the ability to determine how demand effects revenue.
Proof: *Elastic demand means that revenue will fall with increasing prices.*

$$\frac{dR}{dp} = D(p)(1-e), \quad where \;\; e = -p * \frac{D'(p)}{D(p)} \tag{17}$$

Recall revenue which is defined by:

$$R = pq = pD(p)$$

Taking the derivative of revenue R with respect to price p

$$\frac{dR}{dp} = D(p) + pD'(p) = D(p)[1 + p\frac{D'(p)}{D(p)}] \tag{18}$$

We finally prove that:

$$\frac{dR}{dp} = D(p)(1-e) \tag{19}$$

It is now apparent that for values of e *>1*, *(19)* will become negative, thus indicating that forecasting revenue has a strict dependency upon the elasticity of demand.

0.5 Duopolistic Competition

Duopolistic competition characterizes a market scenario dominated by two interdependent firms that wield substantial control over the industry. In this landscape, these two entities hold a significant market share, enabling them to influence prices and, to some extent, each other's strategic decisions. Unlike perfect competition or monopoly, duopolies navigate a complex environment where actions by one firm directly impact the other's market position and profitability. Strategic interactions, such as pricing strategies, product differentiation, and marketing campaigns, become critical tools in their pursuit of market dominance. The equilibrium in a duopoly is delicately balanced between cooperation and competition, with firms vying for market share while considering their competitor's responses. Understanding duopolistic competition sheds light on the nuances of strategic decision-making, market dynamics, and the intricate balance of power within oligopolistic structures.

The Bertrand model presents a strategic framework that delves into pricing dynamics within an oligopolistic market structure. Named after economist Joseph Bertrand, this model departs from traditional assumptions by focusing on firms competing solely on price, assuming homogeneous products and instantaneous adjustments. In this model, firms set prices rather than quantities, aiming to capture the entire market share by undercutting competitors. The equilibrium of the Bertrand model often leads to a situation where prices are driven down to marginal cost, mirroring the competitive outcome of perfect competition despite a few firms dominating the market. This model challenges earlier assumptions about pricing behavior and highlights the pivotal role of price competition in shaping market outcomes.

To further demonstrate the Bertrand model, consider business A and business B. First, business A maximizes revenue from the market by completely positioning itself at the monopolistic price *a1*. This is defined as *e = 1* in *(19)*.

Since Business B believes that business A will hold its production at

$$q_{(A1)} = D(a1), \tag{20}$$

B enters the market by selling at a lower price. The demand curve subject to firm B now becomes:

$$D_{(B1)} = D(p) - q_{(A1)}. \tag{21}$$

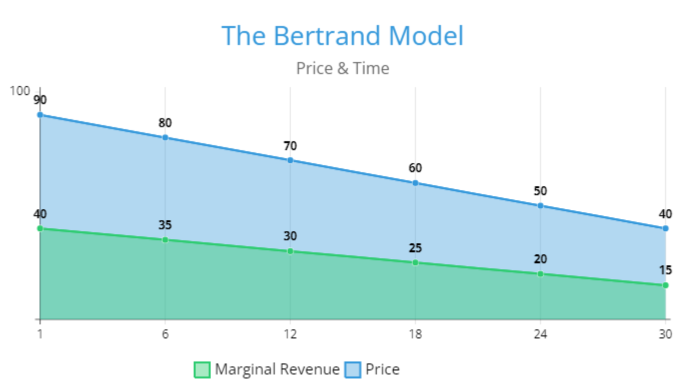
As a result of this, business B positions it's price at $p = b1$. As a response to this change in price, business A responds to it's newly perceived demand:

$$D_{A1}(p) = D(p) - q_{(B1)}. \tag{22}$$

Similar to before, business A now positions it's new price *a2* to be lower than *b1*. From this scenario, we can now see the behaviour of the prices:

$$a1 > b1 > a2 > b2 \ \dots \ a_i > b_i \ \forall \ i \ \in \ Z, \ i > 0 \tag{23}$$

The Bertrand model demonstrates how the undercutting of price based off demand drives the current price of a given commodity towards the marginal cost. Examine the graphic below for a visualized representation of the model.



0.6 Conclusion

The concepts behind elementary microeconomics introduces relations that intricately weave together, forming the foundation upon which modern societies navigate their financial paths. From the foundational principles of supply and demand, revenue, cost, and profit to the complexities of elasticity, duopolistic competition, and pricing models like the Bertrand model, microeconomic theories offer a lens through which we interpret, predict, and guide the behaviors of markets, firms, and consumers. These theories, while diverse in their focus and depth, converge in their pursuit of understanding the human decision-making within the boundaries of resources and incentives. They highlight the balance between rationality and complexity, guiding policies, shaping industries, and shedding light on the ever-evolving dynamics of our economic landscapes. Ultimately, economic theories serve as compasses, navigating the journey toward efficiency, equity, and sustainable prosperity in the global economy.

Bibliography

[1] MacCluer, Charles R. A Survey of Industrial Mathematics. Dover Publications, INC. 2010.